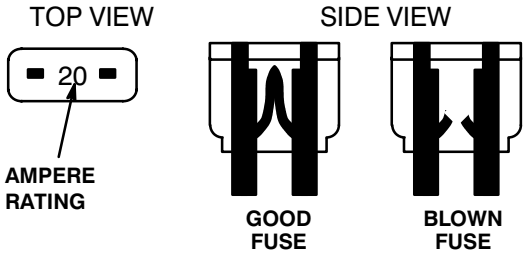
11-1 FUSE PANEL/CIRCUIT PROTECTION

CIRCUIT PROTECTION DEVICES

Electrical circuits on this vehicle may be protected by fuses, fusible links, maxi–fuses, circuit breakers, or a combination of these devices.

BLADE TYPE FUSE

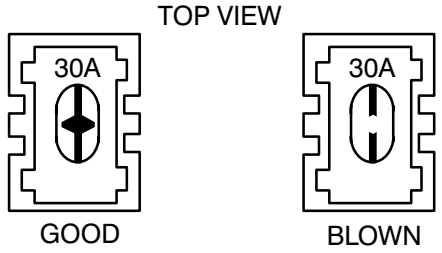


Blade type fuses have a transparent plastic housing. To check a fuse, pull it from the fuse panel and look at the fuse element through the housing. Always replace a blown fuse with a new fuse that has the same ampere rating.

The ampere rating of a blade type fuse can also be determined by following the color code shown here:

BLADE FUSE COLOR CODING	
AMPERE RATING	**HOUSING COLOR**
4	Pink
5	Tan
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

MAXI–FUSE



Cartridge maxi–fuses have transparent plastic tops. To check a cartridge maxi–fuse, look at the fuse element through the top of the housing.

To replace a cartridge maxi–fuse, pull it from the fuse box or panel. Always replace a blown cartridge maxi–fuse with a new one having the same ampere rating.

The ampere rating of a cartridge maxi–fuse can also be determined by following the color code shown here:

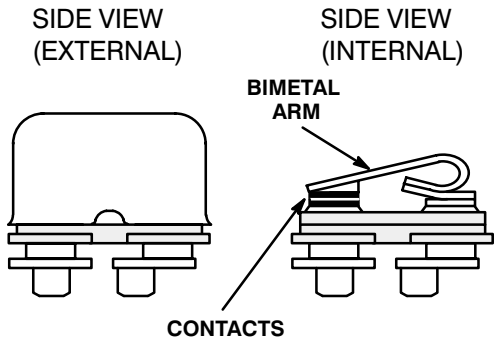
CARTRIDGE TYPE FUSIBLE LINKS	
High Current Fuse Value Amps	**Color Code**
20A Cartridge	Blue
30A Cartridge	Pink
40A Cartridge	Green
60A Cartridge	Yellow

CIRCUIT BREAKER

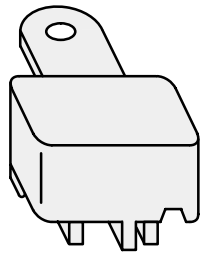
Some circuits are protected by circuit breakers (abbreviated “c. b.” in fuse chart). They can be Fuse Panel mounted or in-line. Like fuses, they are rated in amperes.

Each circuit breaker conducts current through an arm made of two types of metal bonded together (bimetal arm). If the arm starts to carry too much current, it heats up. As one metal expands faster than the other the arm bends, opening the contacts. Current flow is broken. A circuit breaker can be the cycling or non-cycling type.

FUSE PANEL MOUNTED CYCLING TYPE

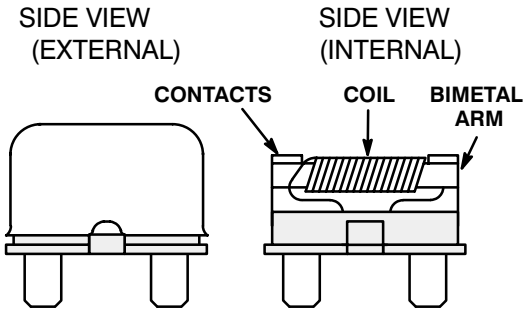


IN–LINE MOUNTED CYCLING TYPE

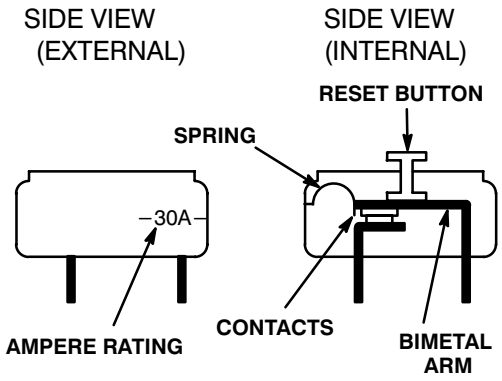


In the cycling type, the bimetal arm cools and straightens out. This cycle repeats as long as the overcurrent exists and power is applied.

FUSE PANEL MOUNTED NON–CYCLING TYPE



FUSE PANEL MOUNTED MANUAL RESET TYPE

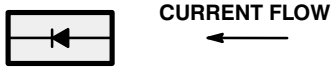


Two types of non–cycling circuit breakers are used; one is reset by removing power from the circuit, and the other is reset by depressing a reset button.

In the first type, there is a coil wrapped around the bimetal arm. When an overcurrent exists and the contacts open, a small current passes through the coil. This current through the coil is not enough to operate a load, but it does heat up both the coil and the bimetal arm. This keeps the arm in the open position until power is removed.

In the second type, a spring pushes the bimetal arm down and holds the contacts together. When an overcurrent condition exists and the bimetal arm heats up, the bimetal arm bends enough to overcome the spring and the contacts snap open. The contacts stay open until the reset button is pushed and the contacts snap together again.

DIODE



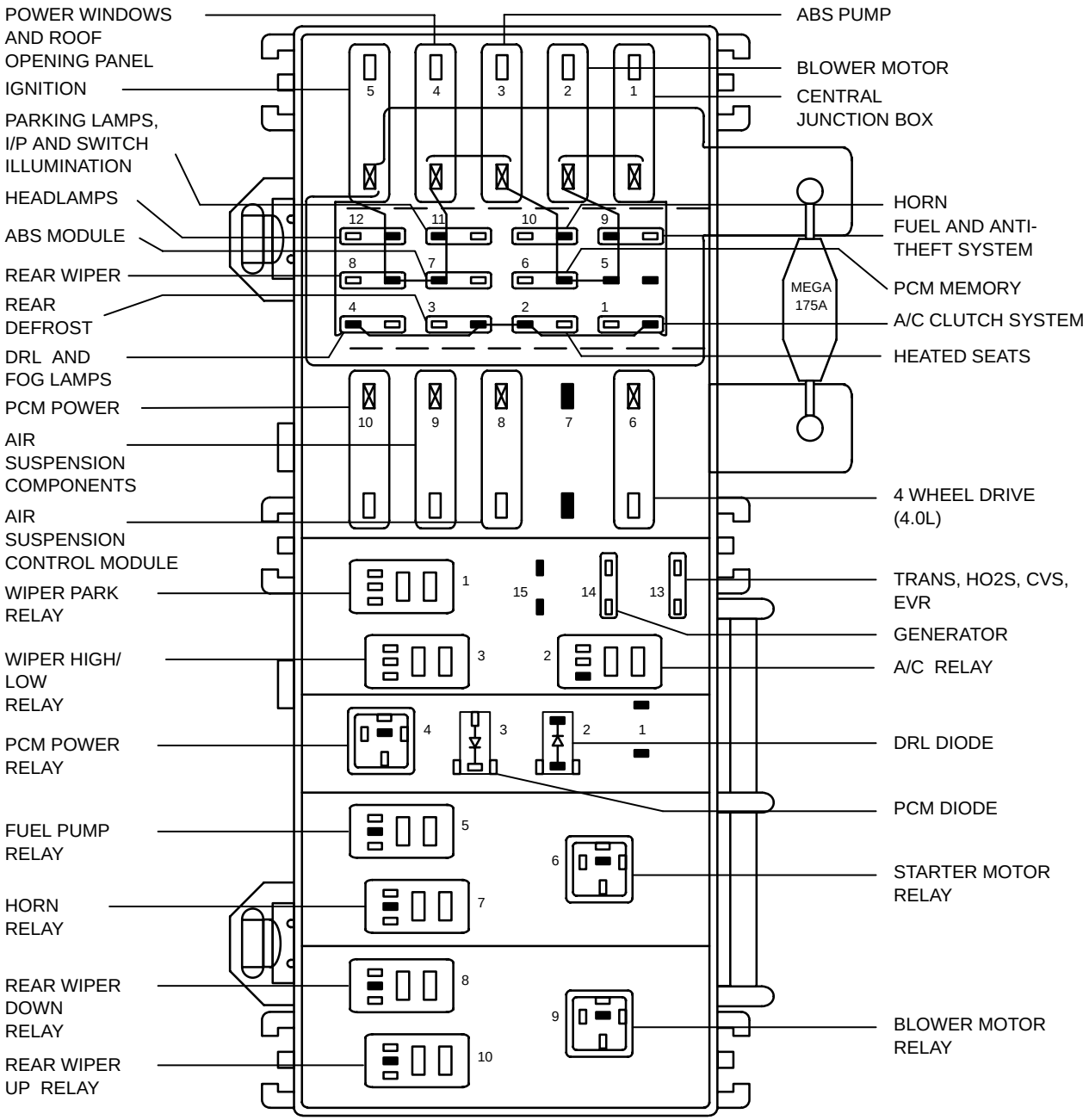
Diodes are electrical devices that permit current to flow in one direction only. The current flows in the direction indicated by the arrow.

MEGA FUSE

TOP VIEW

Mega fuse protects the charging circuit. In the event that the mega fuse is blown due to the charging circuit failure, the generator field circuit is disabled.

11-3 FUSE PANEL/CIRCUIT PROTECTION



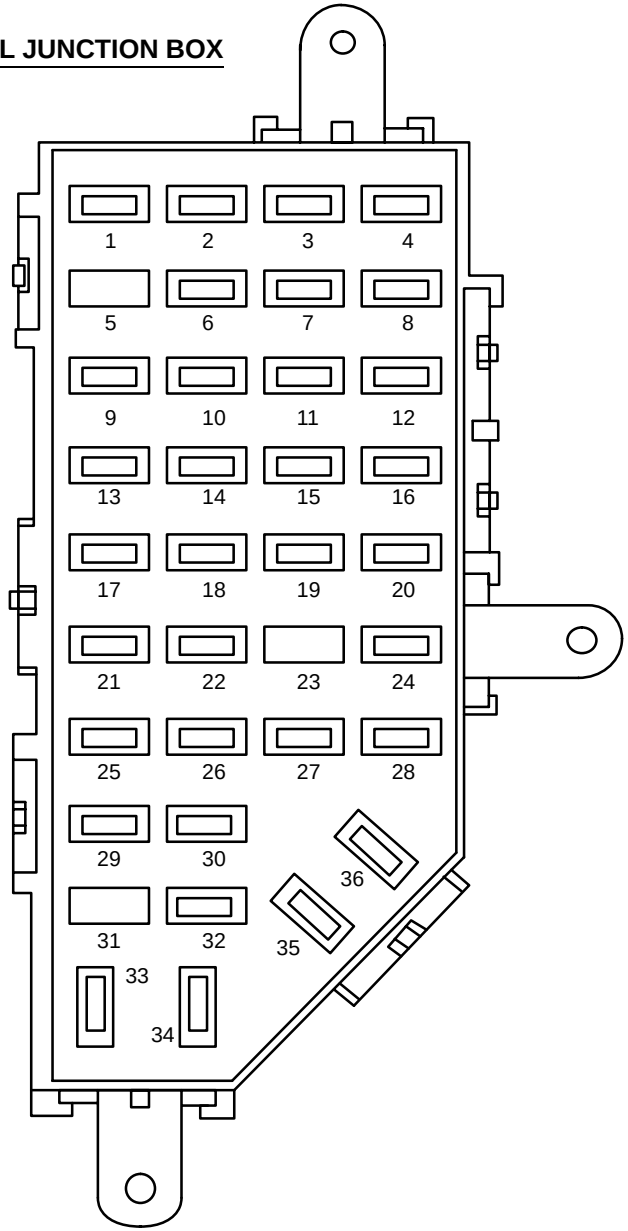
BATTERY JUNCTION BOX

Fuse Position	Fuse/Diode	Amps	Circuits Protected
1	Central Junction Box	60A Maxi	Central Junction Box
2	Blower Motor	40A Maxi	Blower Motor Relay
3	ABS Pump	50A Maxi	4 Wheel Anti-Lock Brake System (4WABS) Module
4	Power Windows and Roof Opening Panel	30A Maxi	Power Roof Opening Panel, Power Windows
5	Ignition	50A Maxi	Ignition Switch, Starter Relay
6	4 Wheel Drive (4.0L)	20A Maxi	Transfer Case Relay
7	–	–	Not Used
8	Adapt. Susp. Mod.	20A Maxi	Air Suspension Control On/Off Switch
9	Adapt.Susp. Comp.	40A Maxi	Air Suspension Control Compressor
10	PCM Power	30A Maxi	PCM Power Relay
1	A/C Clutch System	10A Mini	A/C Clutch
2	Heated Seats	30A Mini	Driver Seat Modules (DSM)
3	Rear Defrost	30A Mini	Rear Window Defrost
4	DRL, Fog Lamps	15A Mini	Fog Lamp, Daytime Running Lamps (DRL)
5	–	–	Not Used
6	PCM Memory	10A Mini	Powertrain Control Module (PCM)
7	ABS Module	30A Mini	4 Wheel Anti-Lock Brake System (4WABS) Module
8	Rear Wiper	15A Mini	Rear Wiper Up Relay, Rear Wiper Down Relay, Rear Wiper Motor
9	Fuel and Anti-Theft	20A Mini	Fuel Pump and RAP Module
10	Horn	15A Mini	Horn Relay, Horn
11	Parking Lamps	15A Mini	Park Lamps Relay, Dimmer Relay, Headlamp Relay, Switch and I/P Illumination, Chime System (Headlamp)
12	Headlamps	30A Mini	Headlamps, Fog Lamp Relay
13	PCM, HO2S, CVS	15A Mini	Heated Oxygen Sensor, EGR Vacuum Regulator, EVR Solenoid, Camshaft Position (CMP) Sensor, Canister Vent Solenoid, Automatic Transmission
14	ALT SYS	30A Mini	Generator
15	–	–	Not Used
1	–	–	Not Used
2	DRL Diode	–	Daytime Running Lamps (DRL)
3	PCM Power Diode	–	Electronic Engine Controls

MAXI-FUSE CARTRIDGE COLOR CODING		MINI FUSE COLOR CODING	
AMPERE RATING	HOUSING COLOR	AMPERE RATING	HOUSING COLOR
20	Yellow	5	Tan
30	Light Green	7.5	Brown
40	Amber	10	Red
50	Red	15	Light Blue
60	Blue	20	Yellow
		25	Natural
		30	Light Green

Relay Position	Relay
1	Wiper Park Relay
2	A/C Relay
3	Wiper High/Low Relay
4	PCM Power Relay
5	Fuel Pump Relay
6	Starter Relay
7	Horn Relay
8	Rear Wiper Down Relay
9	Blower Motor Relay
10	Rear Wiper Up Relay

CENTRAL JUNCTION BOX



FUSE VALUE AMPS	COLOR CODE
5	Tan
7.5	Brown
10	Red
15	Light Blue
20	Yellow
25	Natural
30	Light Green

Fuse Position	Amps	Circuits Protected
1	7.5	Exterior Rear View Mirror Switch, Power Antenna, Memory Seat Switch
2	7.5	Blower Motor Relay, Restraints Control Module (RCM), EATC
3	7.5	Trailer Tow Connector (RH Stop/Turn)
4	10	Left Headlamp
5	–	NOT USED
6	7.5	Rear Blower Motor (W/O EATC)
7	7.5	Trailer Tow Connector (LH Stop/Turn)
8	10	Right Headlamp, Fog Lamp Relay
9	7.5	Brake Pedal Position (BPP) Switch Output
10	7.5	Speed Control Servo, Generic Electronic Module (GEM), Rear Blower Relay (W/EATC), Brake Shift Lock Actuator, Overhead Console, Blend Door Actuator (W/O EATC), Flasher Turn Signals, Air Suspension Control, Steering Rate Sensor, RICP (Rear Blower Switch)
11	7.5	Instrument Cluster, Day/Night Mirror/Autolamp Sensor
12	7.5	Front Washer Control Relay, Rear Washer Control Relay, Washer Pump Motor
13	20	Brake Pedal Position (BPP) Switch, Stoplamps, High Mounted Stoplamps, Brake Pressure Switch (Speed Control), Trailer Tow Stoplamps
14	10	4 Wheel Anti-Lock Brake System (4WABS) Module
15	7.5	Air Bag Indicator Lamp, Charge Indicator Lamp
16	30	Front Wiper Motor, Wiper High/Low Relay, Wiper RUN Relay
17	25	Cigar Lighter, Data Link Connector (DLC)
18	25	Driver Unlock Relay, All-Unlock Relay, All-Lock Relay, Driver and Passenger Power Lumbar, Driver and Passenger Power Seat Motors, Memory Seat Module

Fuse Position	Amps	Circuits Protected
19	25	PCM Power Diode, Ignition Coils, Radio Noise Capacitor, PATS Module, PCM Power Relay
20	7.5	Remote Anti-theft Personality (RAP) Module, Generic Electronic Module (GEM), Radio
21	15	Flasher (Hazard)
22	20	Auxiliary Power Socket
23	-	NOT USED
24	7.5	Clutch Pedal Position (CPP) Switch, Starter Relay Coil, Digital Transmission Range Sensor Switch
25	7.5	Generic Electronic Module (GEM), PATS Module, Speedometer
26	10	Battery Saver Relay, Interior Lamp Relay, Interior Lamps, ACC Delay Relay, Courtesy Lamps
27	15	DRL, Transmission Control Switch (TCS), Rear Window Defrost Relay, Reversing Lamps, E/C Mirror
28	7.5	Generic Electronic Module (GEM), Radio, Memory Seats, Power Antenna
29	25	Radio, Subwoofer, CD Changer, Rear Seat Controls
30	15	Trailer Tow Relay (Parking Lamps)
31	-	NOT USED
32	10	Exterior Rear View Mirror
33	15	High Beam Headlamps, High Beam Indicator, Daytime Running Lamps (DRL) Module
34	7.5	Rear Integrated Control Panel, CD Changer
35	7.5	Rear Blower (W/EATC)
36	7.5	EATC Memory, CD, Rear Integrated Control Panel, Memory Seat, Message Center, Day/Night Mirror/Autolamp Sensor